

COURSE NAME : PRODUCTION ENGINEERING – I
COURSE CODE : 2111
COURSE CATEGORY : E
PERIODS / WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Basic Production Processes: Fitting, Metal Joining Techniques.	15
2	Lathe and Its Operations.	15
3	Shaping, Slotting, Planing Machines.	15
4	Drilling Machines And Its Operations Grinding And Grinding Works	15
Total		60

COURSE OUTCOMES

On completion of the course, students should be able to:

1. Understand the applications of basic production processes.
2. Know the function and working principles of machines like lathe, shaper, slotter, planing and drilling machines.
3. Understand the nomenclature of cutting tools.
4. Materials and bonding process of grinding wheels.

SPECIFIC OUTCOMES

MODULE-I

1.1.0 Understand the areas of application of basic production processes.

- 1.1.1 Identify the fitting tools and their uses.
- 1.1.2 Describe various fitting operations.
- 1.1.3 Explain various metal joining techniques welding, soldering & brazing
- 1.1.4 Define welding process.
- 1.1.5 Classify the welding process and explain its working principle.
- 1.1.6 Illustrate the different arc welding processes. Electrodes, Tools & Equipments.
- 1.1.7 Describe gas welding process, different flames.
- 1.1.8 List the advantages Applications and limitations of welding processes.

- 1.1.9 Identify welding symbols as per BIS.

MODULE-II

2.1.0 Know the working principles of lathe and its uses.

- 2.1.1 Identify the various parts of a Lathe and their functions.
- 2.1.2 Illustrate the dimensional attributes of Lathe.
- 2.1.3. Identify the different work holding and tool holding devices used in a lathe
- 2.1.4. Describe the nomenclature of a single point cutting tool.
- 2.1.5. Define tool angles and choose correct tool angles for a specific work material and working condition.
- 2.1.6 Define speed & feed and select for different work material.
- 2.1.7 Understand the different lathe operations.
- 2.1.8 Describe chip formation.

MODULE-III

3.1.0 Understand the function and working principles of machines like shaping, slotting, planing

- 3.1.1 Illustrate the shaping machine, identify the various parts and their functions.
- 3.1.2 Indicate tool holding and work holding methods in shaping machine
- 3.1.3 Describe the procedure Layout the procedure for cutting grooves, keyways of cutting grooves and key ways etc...
- 3.1.4. Indicate the mode of specification of shaper.
- 3.15 Identify the parts of a slotting machine and their functions.
- 3.1.6 Layout the procedure for cutting keyways, splines etc in a slotting machine.
- 3.1.7 State the working principle of a planer with line sketch.
- 3.1.8 Identify the parts and functions.
- 3.1.9 Indicate the method of fixing work and tool.
- 3.1.10 Identify different types of drilling machine with line sketch.

MODULE-IV

4.1.0 comprehend the function of drilling machine and list the different types of drilling machines

- 4.1.1 Identify the different parts and explain their functions of various drilling machines with line sketch.
- 4.1.2 Explain the work holding and tool holding devices in drilling machines with line sketch.
- 4.1.3 Sketch the drill bit, identify the various parts and angles.
- 4.1.4 List the specification different drilling machines.
- 4.1.5 Identify the various reamers and their uses.
- 4.1.6 Explain the principles of gang and multiple drilling machine.
- 4.1.7 Define grinding process.

- 4.1.8 Understand different materials of grinding wheel.
- 4.1.9 Explain different bonding process.
- 4.1.10 Select grinding wheel and suitable size for a given job.
- 4.1.11 Describe different surface finishing process.

CONTENT OUT LINES

MODULE-I

Introduction about various production processes like metal casting, forging, forming, rolling machines etc.

Fitting:- Definition, Tools: Measuring, Marking, Cutting tools.

Files, Scrapers, Tapes & Dies-Different types and uses.

Fitting Operations: Scribing, punching, chipping, hacksawing, Filing-Different

Filing methods, scraping, drilling, tapping & dieing-hole size.

Metal Joining:-

Different metal joints: Permanent, semi-permanent & Temporary.

Welding:- Definition, Principle, Classification-Fusion, non-Fusion, Different sub classes of welding. Arc welding-principle, types Tool & equipments. Electrodes-types & specifications.

Resistance welding – Principle, Different types (spot, seam, projection, butt and persussion)

Other welding methods:- Thermist, TIG, MIG, Automatic hydrogen welding, ultrasonic welding, plasma arc welding, Electron beam welding and Laser beam welding.

Gas Welding:- Tools, Gases, Different Flames. Gas Cutting Application, Advantages and limitation of welding processes. Welding symbols as per BIS.

MODULE-II

Lathe:- Types-Centre Lathe, Tool Room Lathe, Special purpose lathes. Lathe specification, Lathe parts, Accessories and their functions. Work holding and tool holding devices. Cutting tools: Single point cutting tool-Nomenclature.

Metal Cutting Operations:- Types of chips, Different forces at cutting region. Cutting speed, feed and depth of cut.

Tool life. Factors affecting tool life. Taylor's equation for tool life.

Lathe Operations:- Plain Taper Thread cutting and other operations performed by a lathe.

MODULE-III

Shaper, slotter, planer – General use, Parts & their function, Tool & Die work holding and drill holding devices.

MODULE-IV

Drilling machine and list the different types of drilling machines

Parts and explain their functions of various drilling machines with line sketch.

Work holding and tool holding devices in drilling machines.

Drill bit, identify the various parts and angles.

Specification different drilling machines.

Reamers and their uses. gang and multiple drilling machine

Grinding:- Introduction. Purpose of grinding, Grinding wheels – Abrasive – types – natural & artificial. Types of bonds, Making of grinding wheels: Vitrified, silicate, shellac, Rubber and Bakelite.

Factors affecting selection of grinding wheels, Grade, Grit and structure of grinding wheel. Kind of bond material – wheel shape and sizes. Specification of grinding wheel – Meaning of each letters and numbers.

Other Surface finishing processes:- Introduction. Types of metal finishers.

Electro-plating – Basic principle – plating metals.

Hot dipping – Application, Galvanising.

Tim Coating – Parkerizing, Anodising.

Metal spraying – Wire process, Powder process – Application.

Reference Books:-

1. Production Technology - R K Jain
2. Workshop Technology Vol. I & II - Hajra Chowdary
3. Workshop Technology - W J Chapman
4. Production Technology – HMT